by: Gedalia Koehler

6/21/26

Notes on Matrix Calculus

→ Summary on page 4 !

A few matrix calculus identities — just need to know sums & partial derivatives!

1) $$\boxed{\frac{\partial x_i}{\partial x_j} = \delta_{ij} = \begin{cases} 1, & \text{if } i=j \\ 0, & \text{otherwise.} \end{cases}}$$

2) $$\frac{\partial A\vec{x}}{\partial x_j} = \frac{\partial}{\partial x_j} \sum_i \underbrace{\vec{A}_i \cdot \vec{x}}_{\text{scalar}} \; \underbrace{\hat{i}}_{\text{i hat vector}}$$

$$= \frac{\partial}{\partial x_j} \sum_i \sum_k A_{ik} x_k \hat{i}$$

$$= \sum_i \sum_k A_{ik} \delta_{kj} \hat{i}$$

$$= \sum_i A_{ij} \hat{i} = A^T_j \leftarrow j\text{th column of } A$$

So: $$\boxed{\frac{\partial A\vec{x}}{\partial x_j} = A^T_j} = j\text{th column of } A.$$

What about w.r.t. vector $\vec{x}$?

3) $$\frac{\partial A\vec{x}}{\partial \vec{x}} = \sum_j \frac{\partial}{\partial x_j} A\vec{x} \, \hat{j} = \underbrace{\sum_j A^T_j \, \hat{j}}_{\text{sum over}} = A$$

emphasize that we get new column for each sum.

sum over all columns gets back matrix A.

$$\boxed{\frac{\partial A\vec{x}}{\partial \vec{x}} = A}$$

4) $\dfrac{\partial \vec{x}^T A \vec{x}}{\partial \vec{x}}$. note $\vec{x}^T A \vec{x}$ = scalar $\vec{x}^T \vec{y}$ = scalar
w/ $\vec{y} = A\vec{x}$

$$\vec{x}^T A \vec{x} = \vec{x}^T \sum_k \underset{\uparrow}{A_k} \vec{x}$$

k^{th} row of A

$$= \vec{x}^T \sum_k \sum_i A_{ki} x_i \hat{k}$$

$$= \sum_j x_j \hat{j} \cdot \sum_{ki} A_{ki} x_i \hat{k}$$

$$= \sum_{jki} x_j A_{ki} x_i \underbrace{\hat{j} \cdot \hat{k}}_{\delta_{jk}}$$

$$= \sum_{ki} x_k A_{ki} x_i$$

now use derivative:

$$\frac{\partial \vec{x}^T A \vec{x}}{\partial \vec{x}} = \sum_j \hat{j} \frac{\partial}{\partial x_j} \sum_{ki} x_k A_{ki} x_i$$

use product rule: $\dfrac{\partial}{\partial x_j} x_k \underset{\substack{\uparrow \\ \text{scalar}}}{a} x_i = \left(\dfrac{\partial}{\partial x_j} (x_k a) \right) x_i$

$$+ x_k a \left(\frac{\partial x_i}{\partial x_j} \right)$$

$$= \delta_{kj} a x_i + x_k a \delta_{ij}$$

$$\sum_j \hat{j} \frac{\partial}{\partial x_j} \sum_{ki} x_k A_{ki} x_i$$

$$= \sum_{jki} \hat{j} \left[\delta_{kj} A_{ki} x_i + x_k A_{ki} \delta_{ij} \right]$$

$$= \left[\sum_{ji} \hat{j} A_{ji} x_i \right] + \left[\sum_{jk} \hat{j} x_k A_{kj} \right]$$

sum over columns since summing over outer index $A_{ji} x_i$ = $\vec{x}$ dotted w/ transpose of A = $A^T \vec{x}$

sum over rows = normal matrix multiplication = $A\vec{x}$

$$= A^T \vec{x} + A\vec{x} = (A^T + A)\vec{x}$$

$$\boxed{\frac{\partial \vec{x}^T A \vec{x}}{\partial \vec{x}} = (A^T + A)\vec{x}}$$

5) functions! $f(\vec{x})$ ← takes in vector $\vec{x}$ and returns scalar.

→ use chain rule!

$$\boxed{\frac{\partial f(\vec{x})}{\partial \vec{x}} = \sum_j \hat{j} \frac{\partial}{\partial x_j} f(\vec{x}) = \left[\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \cdots, \frac{\partial f}{\partial x_n} \right]^T}$$

6) products of functions: (product rule ☟)

$$\boxed{\frac{\partial}{\partial \vec{x}} f(\vec{x})\, g(\vec{x}) = \frac{\partial f}{\partial \vec{x}} \cdot g + f \cdot \frac{\partial g}{\partial \vec{x}}}$$

Summary:

1) $$\boxed{\frac{\partial x_i}{\partial x_j} = \delta_{ij} = \begin{cases} 1, & \text{if } i=j \\ 0, & \text{otherwise.} \end{cases}}$$

2) $$\boxed{\frac{\partial A\vec{x}}{\partial x_j} = A^T_j}$$

3) $$\boxed{\frac{\partial A\vec{x}}{\partial \vec{x}} = A}$$

4) $$\boxed{\frac{\partial\, \vec{x}^T A\vec{x}}{\partial \vec{x}} = (A^T + A)\,\vec{x}}$$

5) $$\boxed{\frac{\partial f(\vec{x})}{\partial \vec{x}} = \sum_j \hat{j} \frac{\partial}{\partial x_j} f(\vec{x}) = \left[\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \cdots, \frac{\partial f}{\partial x_n}\right]^T}$$

6) $$\boxed{\frac{\partial}{\partial \vec{x}} f(\vec{x})\, g(\vec{x}) = \frac{\partial f}{\partial \vec{x}} \cdot g + f \cdot \frac{\partial g}{\partial \vec{x}}}$$